Timing a possible encounter

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Separate orbital period from an encounter time.

SUCCESS CRITERIA - I can...

1. Use a clearly stated period and time reference.
2. Check positions of both bodies at the same time.
3. Explain why geometry and uncertainty matter.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “encounter” and “period” to explain your answer. Sentence starters are optional.

encounter: objects reaching nearby positions at the same time

period: time for one complete cycle or orbit

prediction: what a model says may happen next

01 Warm-up: bring back an earlier idea

At fixed mass and speed, circle radius becomes **3** times as large. **Compare acceleration** and net force using equations.

Word help: net force = the vector sum of all forces on one object

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

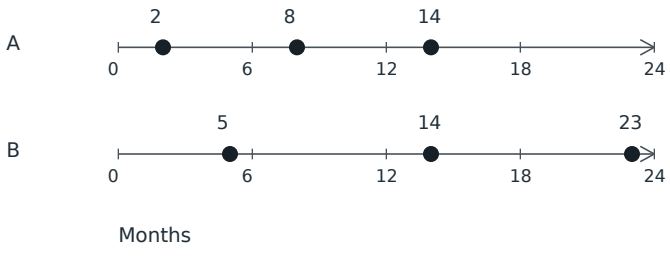
02 Worked example: explain the prediction

Reference: $t_{arrival} = t_{start}$ + travel time.
An encounter needs matching position and time. A period T is one complete orbit. Use the supplied fictional timing information.

Worked reasoning: Two objects pass the same crossing point: A at months **2**, **8**, **14** and B at months **5**, **14**, **23**. Only month **14** is a shared time in these lists.

Your turn: 1. **Draw two timelines** and connect the matching time.
2. **Explain why a shared location** alone is insufficient.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

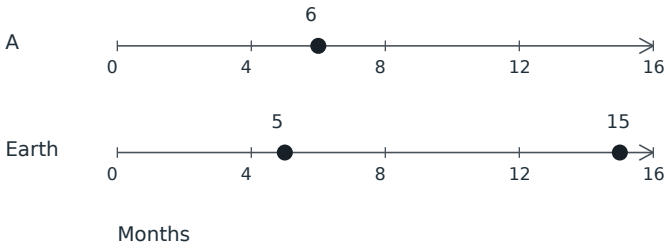


Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

In a simplified orbit example, two paths cross at the same point. Object A arrives in **6 months**. Earth arrives in **5 months** and again in **15 months**. **Mark these times**. Does sharing a crossing point show that the objects will meet? Explain.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

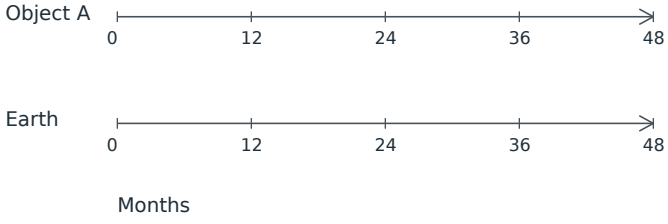


Marzano check - circle: ① ② ③

04 Apply the physics

Toy crossings: asteroid at months **6**, **24**, **42**; Earth at months **6**, **18**, **30**, **42**. **Name common crossing times** and **explain something the prediction leaves** out.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A student turns a ±**2**-day timing range into “a **50%** chance of impact.” Explain why that does not follow.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

An object weighs **39.2 N** where g=**9.8 N/kg**. **Find its mass** and its weight in a field of **6 N/kg**.
Label the masses, field or force relevant to the question.

Word help: weight = the gravitational force on an object

Sentence starter: I found … [unit]. This value means … in this situation.

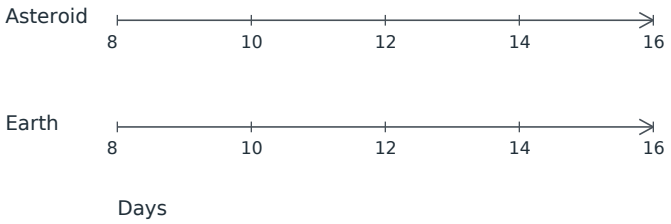


Marzano check - circle: ① ② ③

07 Exit check

An asteroid’s crossing window is days **10**-**14**; Earth’s is days **13**-**15**. **Find overlap** and **state what it** does and does not establish.

Sentence starter: I found … [unit]. This value means … in this situation.

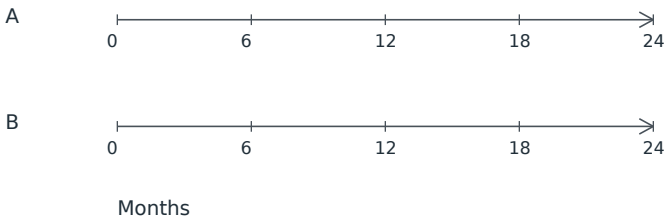


Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A reaches a crossing at months **1**+**6** n; B at months **3**+**8** **m**, for nonnegative integers n, m. **Find their first common crossing** time and something this timing calculation leaves out.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③